

Problem 25.17

If $\delta = 0.06$, calculate the present value of a continuous annuity of 1 payable for 20 years.

Solution

We are given the force of interest

$$\delta = 0.06.$$

The present value of a continuous annuity payable at rate 1 per year for 20 years is

$$\bar{a}_{20|} = \frac{1 - e^{-20\delta}}{\delta}.$$

Substitute $\delta = 0.06$:

$$\bar{a}_{20|} = \frac{1 - e^{-0.06(20)}}{0.06}.$$

Thus,

$$PV = 11.6467479.$$

Therefore, the present value is

$$\boxed{11.647}.$$